Abu Dhabi City Campus Daily/Weekly Planner

Week 9: 26th – 30th October 2026

Subject: Mathematics Diploma Programme **Unit Title: Functions** **Grade: 11**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1 & 2: **Summative** **Number Skills & Sequences** **Financial Applications**	I can demonstrate my understanding of number skills and arithmetic and geometric sequences and series by solving Paper 1 questions accurately and showing clear mathematical working.	Assessment evidence: accuracy, appropriate methods, clear working, mathematical notation and reasonableness of answers.	Lesson notes, assessment instructions and revision resources can be found on Teams and OneNote. Functions, domain, range [IB Maths AI SL/HL] Inverse functions [IB Maths AI SL/HL]
Lesson 3: **Concepts of a Function, Domain & Range**	I can describe a function using notation, graphs, domain and range, and interpret a function as a mathematical model.	2.3 Connect symbolic, graphical and contextual representations of functions. 5.3 Build fluency identifying domain and range from equations and graphs.	
Lesson 4: **Inverse Functions**	I can explain an inverse function as reversing the effect of a function, use the notation $f^{-1}(x)$, and determine or interpret inverse functions when they exist.	2.3 Use mapping diagrams, tables and graphs to show reflection in y = x. 5.3 Practise finding inverse functions and extend checking by composition or substitution.	
Homework / Follow-up	Teams / OneNote: assessment guidance, function and inverse-function notes, targeted practice and feedback.		
Teacher's notes / Reminders	Students must bring a device, graphing calculator/GDC and notebook. GS: Will have one lesson catch up on lesson from Week 8 due to Pink Day celebration		

Technology Integration	Microsoft Teams; graphing calculator/GDC and Desmos for visualising functions, domain/range and reflection in y = x
Cross-curricular links	**Engineering/Science**: functions as models of changing quantities. **Computing:** inverse operations and reversible processes.
External Assessment Integration	International Baccalaureate

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Wellbeing/ ATL Skills	**Thinking:** select appropriate methods, analyse patterns, check reasonableness and connect representations. **Communication:** show clear working and use accurate function notation. **Self-management:** manage time during assessment and reflect on errors. **Social:** discuss and compare strategies during formative function activities.
Standards	IB Mathematics: Applications and Interpretation Functions: Unit 2.2 function concept, notation, domain, range and graphs; inverse functions, one-to-one functions and reflection in y = x.
Inquiry/ TOK Question	How do mathematical models help us make reliable decisions, and what information is needed to reverse a relationship or process? Do you think mathematics or logic should be classified as a language?
Success Criteria (Measurable)	**Lesson 1 & 2: Number Skills & Sequences/Series and Financial Applications Summative** • **All students must** apply number skills and identify/use arithmetic or geometric sequence rules as well as identify the relevant financial information and select an appropriate method for compound interest, depreciation, loans or annuities. • **Most students should** find nth terms and sums and show a clear method for multi-step questions as well as multi-step calculations and interpret the amount, repayment or accumulated value. • **Some students could** solve unfamiliar sequence/series problems and justify the chosen model as well as compare methods or strategies and justify conclusions using mathematical evidence. **Lesson 3 Functions (Domain & Range)** • **All students must** use function notation and identify domain and range from a graph or simple rule. • **Most students should** explain restrictions and connect a function's equation, graph and context. • **Some students could** analyse a less familiar function/model and justify its domain and range. **Lesson 4: Inverse Functions** • **All students must** explain that an inverse reverses a function and recognise $f^{-1}(x)$ notation. • **Most students should** find an inverse for a suitable one-to-one function and verify the relationship using a graph or substitution. • **Some students could** explain why a function must be one-to-one (on the stated domain) and use $f^{-1}(a)$ to solve $f(x)=a$.
	Area: Citizenship / responsible decision-making
	Implementation: Use financial mathematics in the summative assessment to interpret borrowing, saving and depreciation accurately; emphasise checking assumptions and communicating results responsibly.